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Weather conditions and road accidents

Introduction

This project aims to analyze and quantify the relationship between weather conditions and the severity of traffic accidents. While traffic accident data provide valuable information about the circumstances and outcomes of road incidents, they are often insufficient to fully explain observed variations in accident severity. In particular, the influence of external environmental factors, such as weather conditions, remains underexplored when accident data are analyzed in isolation.

Weather conditions can significantly affect road safety by altering visibility, road surface conditions, and driving behavior, thereby increasing the likelihood or severity of accidents. By integrating detailed accident records with historical weather data corresponding to the same time and geographic location, this study addresses this limitation and provides a more comprehensive framework for accident analysis. Understanding the relationship between meteorological conditions and accident severity is essential for improving risk assessment, informing preventive measures, and supporting data-driven road safety policies.

The resulting dataset enables exploratory data analysis, statistical correlation studies, and predictive modeling, with the objective of identifying weather patterns that may increase the risk or severity of traffic accidents.

1. Data Collection and Aggregation

To address the objectives of the project, two main categories of data were collected: accident-related data and historical weather data. The data acquisition strategy combined web scraping and static dataset collection, chosen according to data availability, reliability, and project constraints.

The Accident data was obtained from a publicly available dataset provided by the French government in CSV format. This dataset contains detailed records of traffic accidents, including temporal, spatial, and contextual information.

Weather data were collected using two complementary sources. First, web scraping was performed using Python and Selenium to retrieve historical weather conditions corresponding to the time and location of individual accidents from the Wunderground website. However, an initial analysis revealed that certain meteorological variables were not available at the required level of completeness or consistency. To address this limitation, a third data source was incorporated: a historical weather dataset provided by the French government, which was used to supplement and complete the weather information.

Following data acquisition, the collected datasets were analyzed to identify relevant features for modeling. A preprocessing phase was then applied to clean, transform, and integrate the data into a unified dataset suitable for subsequent analysis and model training.

1.1 Data Sources

At the planning stages of this part, we targeted two main sources of data. One of the sources was the direct acquisition of the CSV of the accidents from the French Government's website (found here).

The second data source consists of historical weather conditions corresponding to the time and location of each accident. WeatherUnderground was initially selected because it provides weather information with sufficient spatial and temporal granularity, which is essential for accurately matching meteorological conditions to individual accidents. Potential challenges associated with web scraping, including access restrictions and the time required to collect a large volume of data (over 100,000 records), were considered in advance. WeatherUnderground was found to allow sustained data collection with minimal blocking, making the data acquisition process feasible despite the extended collection time required.

However, after having done the data acquisition of both datasets and then analyzing them, we found that while the government-issued dataset of the accidents proved sufficient for our needs, the WeatherUnderground acquired dataset still has some issues. There was a substantial amount of missing data in the result, data that we had determined to be important enough to not merely filter. We surmised that this is the case because there is a good amount of weather stations, the sensor that collects the data of the weather, that outright do not track the data that we want, such as visibility and ground conditions. Thus, we decided to add another data source to supplement the missing data.

The supplement that we chose comes from the weather history data from the French Government themselves (found here). We have found that the data characteristics mesh well with the characteristics of the WeatherUnderground dataset, which is that it is constrained in the point of time and space. The dataset comes in the form of multiple CSVs of each district that each contain hourly weather data of a weather station in the France region.

1.2 Data Collection Process

For this section, we will focus on the last two collection processes, as the collection process of the CSV of the accident we consider to be straightforward. Furthermore, we have provided a GitHub link to the code of our scripts along with some preliminary results from the data acquisition (found here).

For the process of web scraping the weather data, we wrote a python script to web-scrap the WeatherUnderground website. Accident location (latitude and longitude) and time information were extracted from the accident dataset. A REST API was used to convert geographic coordinates into city names, which were then used to construct time-specific URLs for web scraping historical weather data from WeatherUnderground using Selenium. The retrieved weather data were stored in a CSV file together with the corresponding accident identifiers, enabling subsequent dataset merging.

For the second data collection process, accident data were obtained from the CSV files provided by the French government. However, the large volume of the dataset, exceeding 2

GB in compressed form and over 20 GB once uncompressed, made direct processing using standard Python-based tools impractical.

To address this limitation, the data were imported into a PostgreSQL database. This approach enabled efficient indexing and significantly improved query performance, reducing execution times from several minutes to milliseconds. In particular, the use of the PostGIS extension allowed for efficient spatial and temporal queries, which were essential for matching accident locations with corresponding weather data.

Hourly historical weather data were obtained from the French government's open data platform and stored in the same database. A custom processing script was developed to associate each accident with the relevant weather conditions based on geographic coordinates and time of occurrence. The resulting enriched dataset was then exported to a CSV file, forming the final output of the data collection stage.

As the last step of the process, we aggregated all three results into a single aggregated dataset. Thus, we have a dataset shaped (110124, 33). The features that are in the dataset is:

['Num_Acc', 'jour', 'mois', 'an', 'hrmn', 'lat', 'long', 'com', 'dep', 'lum', 'atm', 'agg', 'col', 'adr', 'catr', 'circ', 'surf', 'infra', 'situ', 'plan', 'prof', 'vma', 'nb_vehicules', 'has_2RM', 'has_VL', 'has_PL', 'has_TC', 'has_MD', 'has_OTHER', 'choc_multiple', 'nb_usagers', 'grav_max', 'age_moyen']

With the combined dataset, we started to measure the quality of our data for preprocessing

2. Data Preprocessing

We have done a quality assessment of our data, with the results as described here. From the gathered dataset, we first checked for any null or incomplete values. In brief, there were the features that we found to have null values.

- Visibility: 32832 of the data missing, 32% of total data.
- Wind_Speed: 3226 of the data missing, 2.93% of the total data.
- Temperature: 3226 of the data missing, 2.93% of the total data.
- Pressure: 3226 of the data missing, 2.93% of the total data.
- Circ: 7046 of the data missing, 7% of the total data.
- Vma: 1910 of the data missing, 1.73% of the total data.
- Infra: 1200 of the data missing, 1.09% of the total data.
- Col: 76 of the data missing, 0.07% of the total data.
- Situ: 63 of the data missing, 0.06% of the total data.
- Prof: 61 of the data missing, 0.06% of the total data.
- Surf: 60 of the data missing, 0.06% of the total data.
- Atm: 5 of the data missing, 0.004% of the total data.
- Lum: 4 of the data missing, 0.004% of the total data.

These missing features will need to be handled. The steps of the handling that we've done will be in the pre-treatment phase.

For the pre-treatment of the dataset, the main part of the process is data cleaning. It is to note that the Python Notebook that we used in this process is also in the GitHub repository (here). The data cleaning that we've done can be generally divided into a couple of steps.

The first few of the steps are fundamental. We changed the -1.0 value from the acquired data into NaN (not a number), and then changed the NaN in a number variable into the string 'inconnu' in order to be uniform with the other types of variable. Then, we filled missing speed limits using a conditional median based on the road category.

2.1 Outlier Handling and Robustness Strategy

To address extreme values in numerical variables, a quantile-based outlier handling strategy was adopted. For each relevant variable, appropriate lower and upper bounds were selected based on empirical quantiles, taking into account both the proportion of affected observations and the interpretability of extreme values from a domain perspective.

Rather than removing outliers, a clipping strategy was applied in order to preserve the full dataset while limiting the influence of extreme observations on model training. Values exceeding the upper bound were replaced by the median of the observations above this threshold, while values below the lower bound were replaced by the median (or mean when more appropriate) of the observations below the corresponding lower threshold.

This approach makes it possible to reduce the impact of extreme values without discarding data, while still retaining information about whether an observation originally lay above or below the selected bounds. By correcting extreme values in a controlled and interpretable manner, the resulting dataset remains robust, statistically consistent, and well suited for downstream modeling, without introducing artificial truncation effects or reducing sample size.

2.2 Categorical Variable Transformation

Many variables in the BAAC accident dataset, such as circ, infra, col, situ, prof, surf, and plan, are categorical by nature, but are originally encoded using numerical codes. Each numerical value corresponds to a specific semantic meaning defined in the official BAAC documentation.

As part of the preprocessing, these variables were converted into explicit categorical features, by replacing each numerical code with its corresponding descriptive label. This transformation improves readability, interpretability, and facilitates model analysis.

In addition, the dimensionality of several categorical variables was deliberately reduced by merging semantically close categories. The objective was to simplify the feature space while preserving the essential information. For example, the variable *surf*, which originally contained seven distinct categories describing road surface conditions, was regrouped into four broader and more interpretable categories: *normal*, *wet*, *slippery*, and *degraded*.

For each categorical variable, missing or invalid values (including placeholder codes such as -1) were handled consistently. When missing information was frequent or could not be

reliably inferred, it was assigned to an explicit *"unknown"* category (*inconnu*), ensuring that uncertainty was preserved rather than removed.

Overall, this approach allowed us to retain the qualitative nature of these variables, improve interpretability, reduce sparsity, and prepare the dataset for effective modeling.

3. Modeling and analysis of the influence of weather on accident severity

After presenting the data collection and aggregation process, as well as the different preprocessing steps, this third section is dedicated to modeling. The main objective is to train classification models capable of predicting the maximum severity of an accident (grav_max) based on the available characteristics, and to analyze these results in relation to our research question.

In a first step, models are trained without using meteorological variables, relying only on information describing the accident, road infrastructure, temporal context, and the users involved. This step makes it possible to establish a baseline that provides an estimate of the models' performance in the complete absence of weather information. In a second step, the same models are trained by integrating weather data, in order to assess whether these additional variables improve overall performance or the prediction of certain severity classes, particularly the most severe accidents.

Finally, beyond raw performance metrics, a model interpretability analysis is conducted. Using methods such as feature importance or Partial Dependence Plots (PDP), the specific impact of meteorological variables on accident severity is examined, making it possible to go beyond a simple score comparison and to provide concrete interpretative insights.

3.1 Models without weather data

3.1.1 Target variable and class imbalance

The target variable used is the maximum accident severity (grav_max), defined according to three classes:
Class 2: fatal accident,
Class 3: hospitalized injury,
Class 4: slight injury.

The distribution of this variable is highly imbalanced:
Class 4 (slight injury): around 70% of the observations,
Class 3 (hospitalized): around 25%,
Class 2 (fatal): around 4%.

We are therefore dealing with a class imbalance context, which is not abnormal as it reflects the reality of accident data, where fatal cases remain rare. However, it makes the classification task more complex, especially for class 2, whose decision boundary is more difficult to learn. We can already assume that accuracy alone is not sufficient to evaluate model performance, since a model that always predicts class 4 would already achieve 70% accuracy. Therefore, metrics that take class balance into account should be favored.

3.1.2 Explanatory variables used

We reduce the dataset to variables related to the temporal context (day, month, year, hour), lighting and traffic conditions, road infrastructure characteristics (road type, layout, profile, surface, intersection, situation), involved vehicles (number of vehicles, presence of powered two-wheelers, heavy goods vehicles, public transport, etc.), users (number of users, average age), and crash dynamics (multiple collision or not).

No weather-related information is included at this stage, which allows us to isolate the effect of structural accident characteristics only.

3.1.3 Models and training strategy

Two types of tree-based models were selected:
- Decision Tree (hyperparameter search on max_depth and min_samples_leaf),
- Random Forest.

These models have the advantage of being well suited to heterogeneous data (numerical and encoded categorical variables) and of allowing interpretability analysis at a later stage.

For each type of model, a hyperparameter search was conducted to optimize performance. For the Decision Tree, the search focused on max_depth and min_samples_leaf. For the Random Forest, two hyperparameter searches were carried out. A “simple” one focusing on n_estimators, max_depth, and min_samples_leaf, and a “complex” one additionally including min_samples_split and max_features.

The models were selected based on two distinct criteria: accuracy, which favors overall performance, and macro F1-score, which is more appropriate for class imbalance as it gives equal weight to each class. This approach leads to the evaluation of several model variants, allowing comparison of the impact of hyperparameter choices and optimization metrics.

3.1.4 Results and performance analysis

Modele	Accuracy	Precision_macro	Recall_macro	F1_macro	Precision_C2	Recall_C2	Precision_C3	Recall_C3	Precision_C4	Recall_C4
DT (F1)	0.623	0.451	0.529	0.447	0.137	0.538	0.402	0.306	0.814	0.742
DT (Acc)	0.601	0.443	0.531	0.433	0.124	0.594	0.379	0.284	0.827	0.716
RF (F1 complex)	0.672	0.49	0.528	0.504	0.219	0.343	0.426	0.478	0.826	0.762
RF (Acc complex)	0.672	0.49	0.528	0.504	0.219	0.343	0.426	0.478	0.826	0.762
RF (F1 simple)	0.649	0.472	0.537	0.485	0.181	0.443	0.405	0.431	0.829	0.739
RF (Acc simple)	0.616	0.448	0.536	0.448	0.143	0.564	0.376	0.317	0.824	0.727

The overall table presented earlier compares six models trained without meteorological data: two Decision Tree models (optimized for accuracy or macro F1-score) and four Random Forest models, resulting from “simple” and “complex” hyperparameter searches, also optimized according to these two criteria.

It should be noted that the Random Forest models obtained from the complex hyperparameter search, optimized respectively for accuracy and for macro F1-score, exhibit exactly the same performance across all metrics. This indicates that, in this configuration,

the search converged toward the same set of hyperparameters, regardless of the optimization metric used.

Overall performance analysis (macro averages and accuracy)

On global metrics, Random Forest models clearly outperform Decision Trees.

The best overall results are obtained by Random Forest models optimized either on accuracy or on macro F1-score:

- Accuracy = 0.672,
- Macro F1-score = 0.504,
- Macro precision = 0.490,
- Macro recall = 0.528.

Decision Trees show more limited performance:

- DT(F1): accuracy 0.623, macro F1-score 0.447,
- DT(Acc): accuracy 0.601, macro F1-score 0.433.

A first conclusion can be drawn: without meteorological data, a model based solely on accident characteristics, infrastructure, and users reaches around 0.67 accuracy, but more importantly a macro F1-score of about 0.50. This reflects a complex classification task, marked by strong class imbalance and decision boundaries that are difficult to learn.

Why accuracy is not sufficient in this context

Since the target variable grav_max is highly imbalanced, a good accuracy may simply reflect a good prediction of the majority class (slight injury), without guaranteeing satisfactory detection of the most severe accidents.

It is therefore essential to complement this global reading with a class-by-class analysis, relying on precision, recall, and F1-score metrics, particularly for class 2 (fatal accidents).

Class-wise analysis: what the models actually do

Class 4: Slight injury

The majority class is correctly predicted by all models. The best performance is achieved by Random Forest models, with a precision of 0.826 and a recall of 0.762, corresponding to an F1-score of approximately 0.79.

This result is expected, as this class represents the majority of observations, which facilitates the learning of its patterns.

Class 3: Hospitalized injury

It is in this intermediate class that Random Forest models provide a significant gain. The optimized models reach:

- Precision = 0.426,
- Recall = 0.478,
- F1-score ≈ 0.45,

Compared to clearly lower performance for Decision Trees, this improvement suggests that Random Forests more effectively capture the combination of factors describing accidents located at the boundary between slight accidents and more severe accidents requiring hospitalization.

Class 2: Fatal accident: revention vs false alerts trade-off

The class of fatal accidents represents the main difficulty.
The Decision Tree optimized on accuracy maximizes recall:

- Recall_C2 = 0.594, meaning nearly 60% of fatalities are detected,
- but Precision_C2 = 0.124, indicating a very high number of false positives.

This model follows a prevention-oriented logic, at the cost of a large number of false alerts.

Conversely, Random Forest models optimized on macro F1-score favor a more balanced compromise:

- Precision_C2 = 0.219,
- Recall_C2 = 0.343,

which leads to the best F1-score for class 2. These models detect fewer fatal accidents, but their predictions are more reliable when a fatal outcome is predicted.

Finally, Random Forest models from the simple hyperparameter search can reach high recall values, but at the cost of very low precision, once again illustrating the trade-off between maximum detection and prediction reliability.

Summary and transition

This analysis highlights that, without meteorological data, the models are already able to capture a significant share of the information explaining accident severity, particularly thanks to Random Forest models. However, the prediction of fatal accidents remains difficult and strongly depends on the trade-off between recall and precision.

These results constitute a solid baseline, from which it becomes possible to assess the contribution of additional variables. The following section builds on this reference in order to analyze whether the integration of meteorological data improves overall model performance or modifies model behavior, particularly for the detection of the most severe accidents.

3.2 Models trained with meteorological data

After establishing a reference based on models trained without meteorological data, this second subsection aims to evaluate the contribution of weather variables in predicting accident severity. The objective is to determine whether integrating weather data improves overall model performance and to analyze whether these variables contribute more specifically to better detection of the most severe accidents, particularly fatal accidents.

3.2.1 Integration of meteorological variables

In this configuration, the dataset used is identical to that of section 3.1, with the addition of meteorological variables associated with each accident.

These variables describe atmospheric conditions at the time of the accident, including weather conditions, precipitation, temperature, visibility, and the presence of specific conditions (rain, snow, fog, etc.).

3.2.2 Target variable and class imbalance

The target variable remains unchanged: it is still the maximum accident severity (grav_max), and the class distribution is identical to that observed in the previous section. The class imbalance issue therefore remains. As a result, the same methodological precautions are maintained. Accuracy alone is not sufficient to evaluate the models, and particular attention is paid to class-wise metrics, especially recall and precision for class 2 (fatal accidents). This ensures a consistent comparison between models with and without weather data.

3.2.3 Models and training strategy

The models used with meteorological data are strictly the same as those employed in section 3.1, with the same hyperparameter searches. This methodological consistency ensures that any observed variation in performance can be attributed to the addition of meteorological variables, rather than to a change in model or optimization strategy (same random seeds).

3.2.4 Results and performance analysis

model	accuracy	precision_macro	recall_macro	f1_macro	precision_2	recall_2	f1_2	precision_3	recall_3	f1_3	precision_4	recall_4	f1_4
DT F1	0.620	0.450	0.530	0.447	0.137	0.547	0.219	0.399	0.307	0.347	0.814	0.737	0.774
DT ACC	0.600	0.446	0.534	0.433	0.122	0.605	0.203	0.390	0.283	0.328	0.827	0.714	0.766
RF F1	0.682	0.493	0.512	0.500	0.220	0.273	0.244	0.434	0.485	0.458	0.824	0.777	0.800
DT ACC	0.684	0.496	0.515	0.504	0.224	0.278	0.248	0.438	0.487	0.461	0.824	0.780	0.801

The overall table presented earlier compares four models trained with meteorological data: two Decision Tree models (optimized for accuracy or macro F1-score) and two Random Forest models (optimized according to the same criteria).

Overall performance analysis (macro averages and accuracy) and comparison with the baseline:

On global metrics, Random Forest models clearly outperform Decision Trees, confirming the observations made without meteorological data.
The best overall results are obtained by Random Forest models integrating weather data:

- Accuracy = 0.684 (vs. 0.672 without weather),
- Macro F1-score = 0.504 (identical to the configuration without weather),
- Macro precision = 0.496 (vs. 0.490),
- Macro recall = 0.515 (vs. 0.528).

Decision Trees remain less competitive:

- DT(F1): accuracy 0.620 (vs. 0.623), macro F1-score 0.447 (identical),
- DT(Acc): accuracy 0.600 (vs. 0.601), macro F1-score 0.433 (identical).

A first conclusion can be drawn: integrating weather data leads to a slight gain in overall accuracy, but does not improve the macro F1-score. This suggests that the global balance between classes remains similar to what was observed without meteorological data.

Class-wise analysis: what the models actually do

Class 4: Slight injury

The majority class is correctly predicted by all models. The best performance is achieved by Random Forest models, with a precision of 0.824 and a recall close to 0.78 (F1 ≈ 0.80).
Even for this dominant class, Random Forest models show a better precision/recall balance than Decision Trees. The addition of weather data leads to a slight improvement in performance, indicating a marginal but consistent contribution of these variables.

Class 3: Hospitalized injury

The intermediate class benefits more clearly from the use of Random Forest models. The model optimized on accuracy reaches:

- Precision = 0.438 (vs. 0.426 without weather),
- Recall = 0.485 (vs. 0.478).

Decision Trees show more limited changes. These results confirm the ability of Random Forest models to better capture the transition between slight accidents and severe accidents, and suggest that weather data slightly contribute to this differentiation.

Class 2: Fatal accident, prevention vs false alerts trade-off

Class 2 remains the most difficult to predict.
The Decision Tree optimized on accuracy maximizes recall:

- Recall_C2 = 0.547 (vs. 0.594 without weather),
- but Precision_C2 = 0.122 (vs. 0.124).

This model follows a prevention-oriented logic, detecting a large share of fatal accidents at the cost of many false positives.

Conversely, the Random Forest optimized on accuracy with weather data prioritizes reliability:

- Precision_C2 = 0.224 (vs. 0.219 without weather),
- Recall_C2 = 0.278 (vs. 0.343).

The addition of weather data therefore makes the model more conservative: it predicts fewer fatal accidents, but its predictions are slightly more reliable when a fatal outcome is announced.

Summary

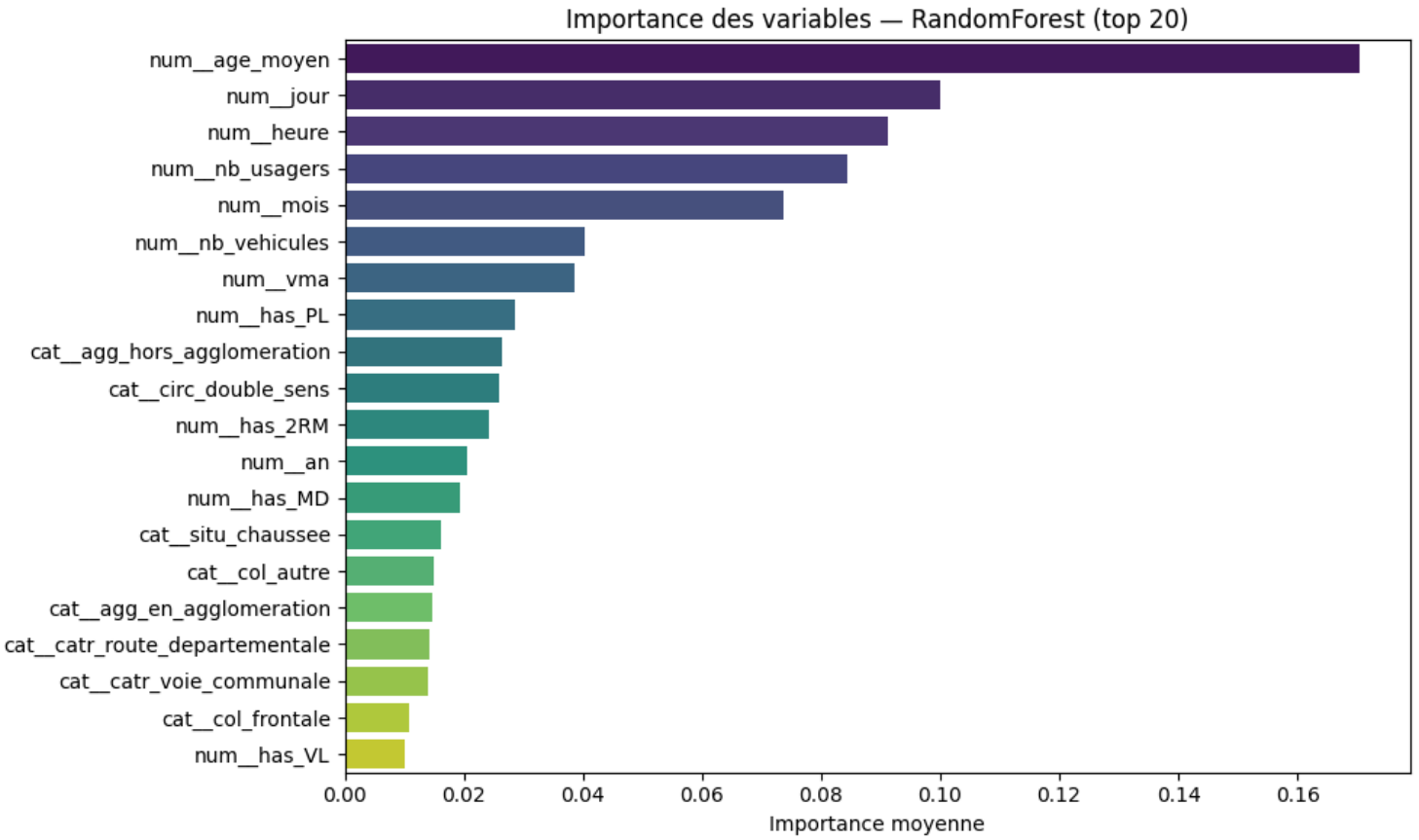
Integrating meteorological data leads to slight but consistent improvements for classes 3 and 4, and modifies the precision/recall trade-off for class 2. While weather data do not significantly increase the detection of fatal accidents, they nevertheless act as a secondary discriminating factor, improving the reliability of certain predictions without fundamentally transforming overall performance. These results justify the interpretability analysis presented in the following section, in order to better understand the specific influence of meteorological variables on accident severity.

3.3 Model interpretability

After analyzing the performance of models trained with and without meteorological data, it is necessary to go beyond global metrics in order to better understand how these models make their decisions. Indeed, similar performance levels may result from very different internal mechanisms, and the analysis of feature importance provides additional insight into model behavior.

This section is dedicated to model interpretability, with the objective of identifying the most influential variables in predicting accident severity and assessing the relative role of different categories of variables. Particular attention is given to meteorological variables.

3.3.1 Visualization of models without weather data



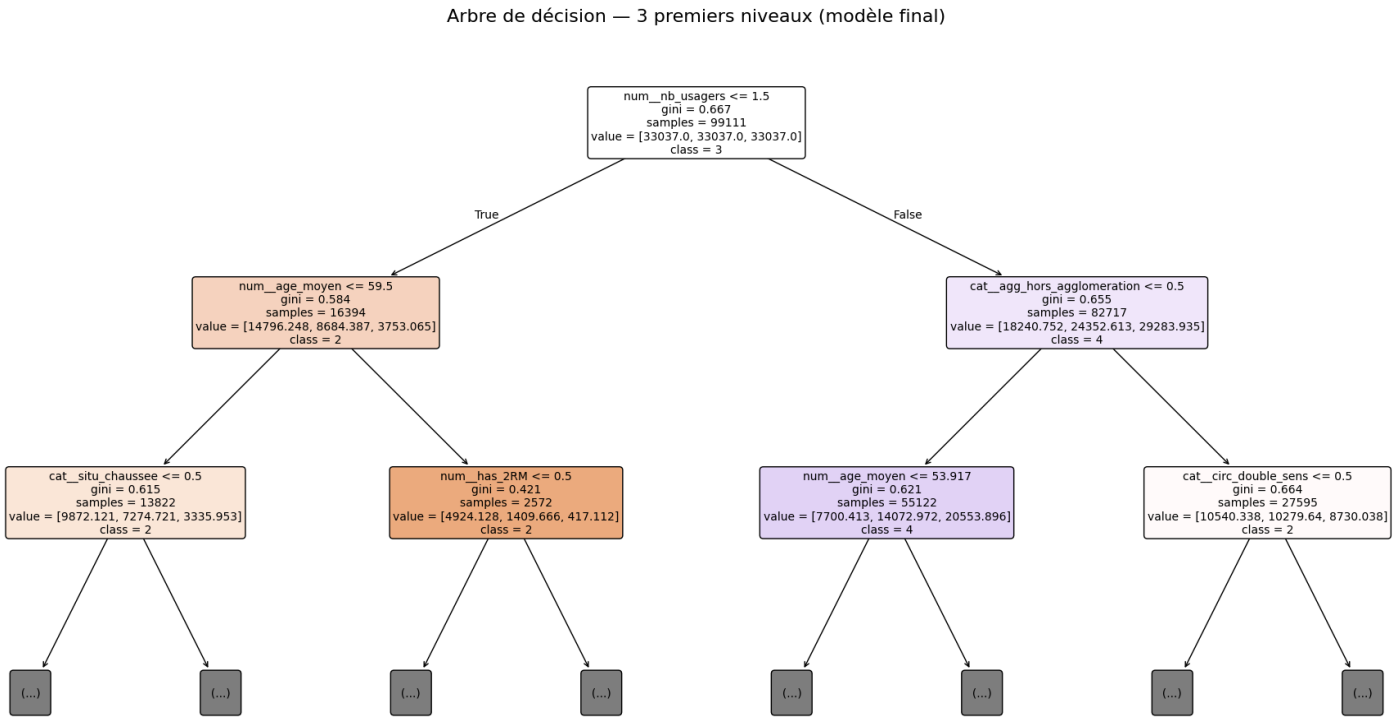
Feature importance Random Forest

The analysis of feature importances derived from the Random Forest (F1 complex) highlights the predominant role of the average age of users, which appears as the most influential variable in predicting accident severity. This observation emphasizes the importance of the vulnerability of the individuals involved. Temporal factors also occupy a

central position, with high importance associated with day, hour, and month. These variables reflect risk exposure effects, related to traffic conditions, peak periods, and differentiated behaviors depending on the time of day or year.

Next comes variables describing the configuration of the accident and the traffic context, such as the number of users involved, the number of vehicles, and the maximum authorized speed. These factors reflect both the potential intensity of the impact and the complexity of the accident situation.

Finally, certain infrastructural characteristics (road type, position on the roadway, two-way traffic, location within or outside urban areas) appear as complementary variables, playing a secondary but non-negligible role in modeling accident severity.



Interpretation of the decision tree (DT F1)

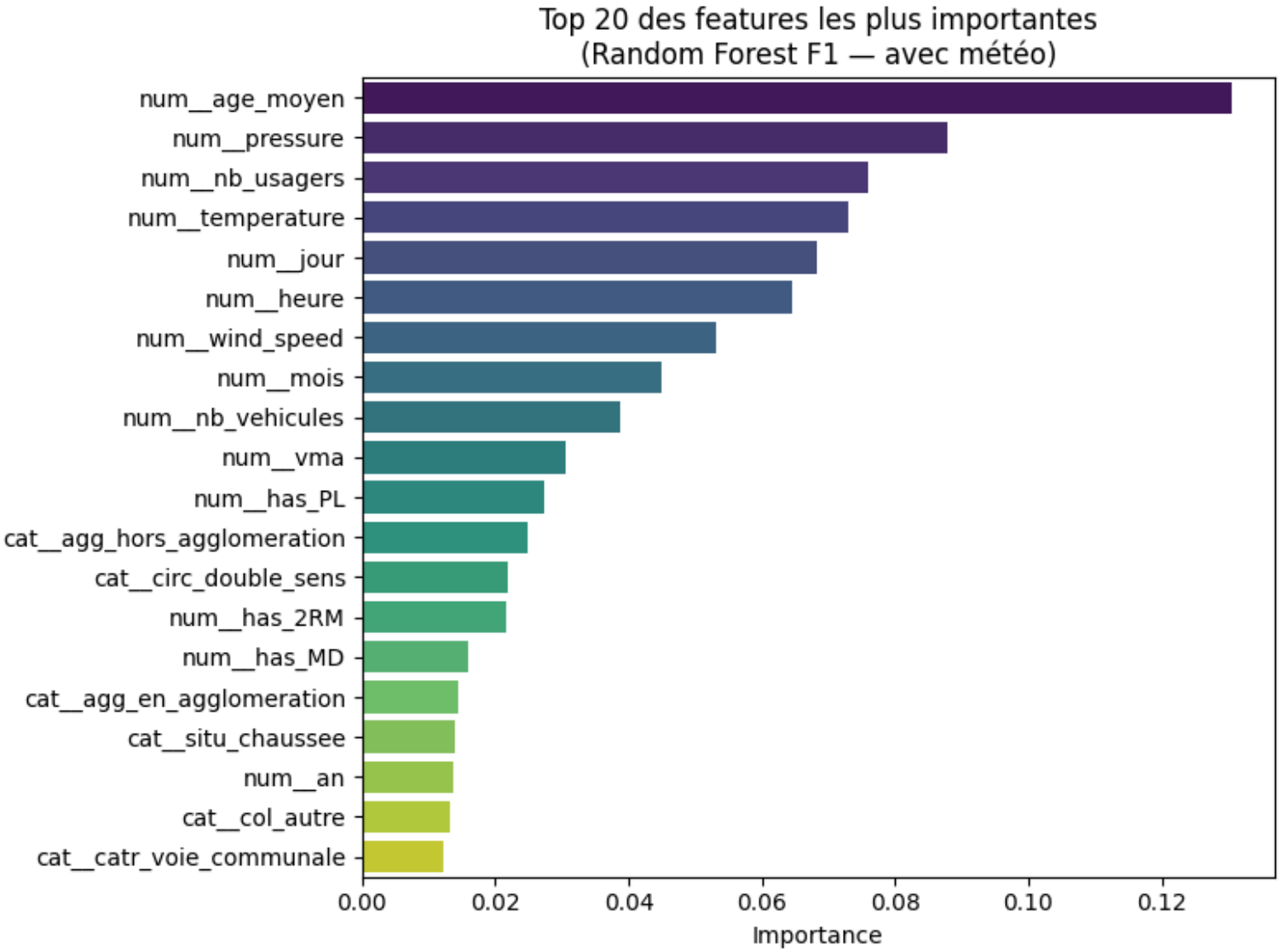
The visualization of the first levels of the decision tree confirms these observations. From the root node, the number of users involved constitutes the first splitting criterion, indicating that it is a structuring factor for severity prediction. At the subsequent levels, the average age of users appears repeatedly, as do certain variables related to road configuration and traffic conditions.

This hierarchy of variables within the tree provides an intuitive illustration of the model's decision logic: accidents involving multiple users combined with greater vulnerability are associated with higher accident severity.

Consistency between models

The comparison between the Decision Tree and the Random Forest highlights a strong consistency in the variables identified as discriminative. The dominant factors, the number of users involved, average age, temporal context, and general road characteristics, appear both in the early splits of the tree and among the most important variables in the forest.

3.3.2 Visualization of models with weather data

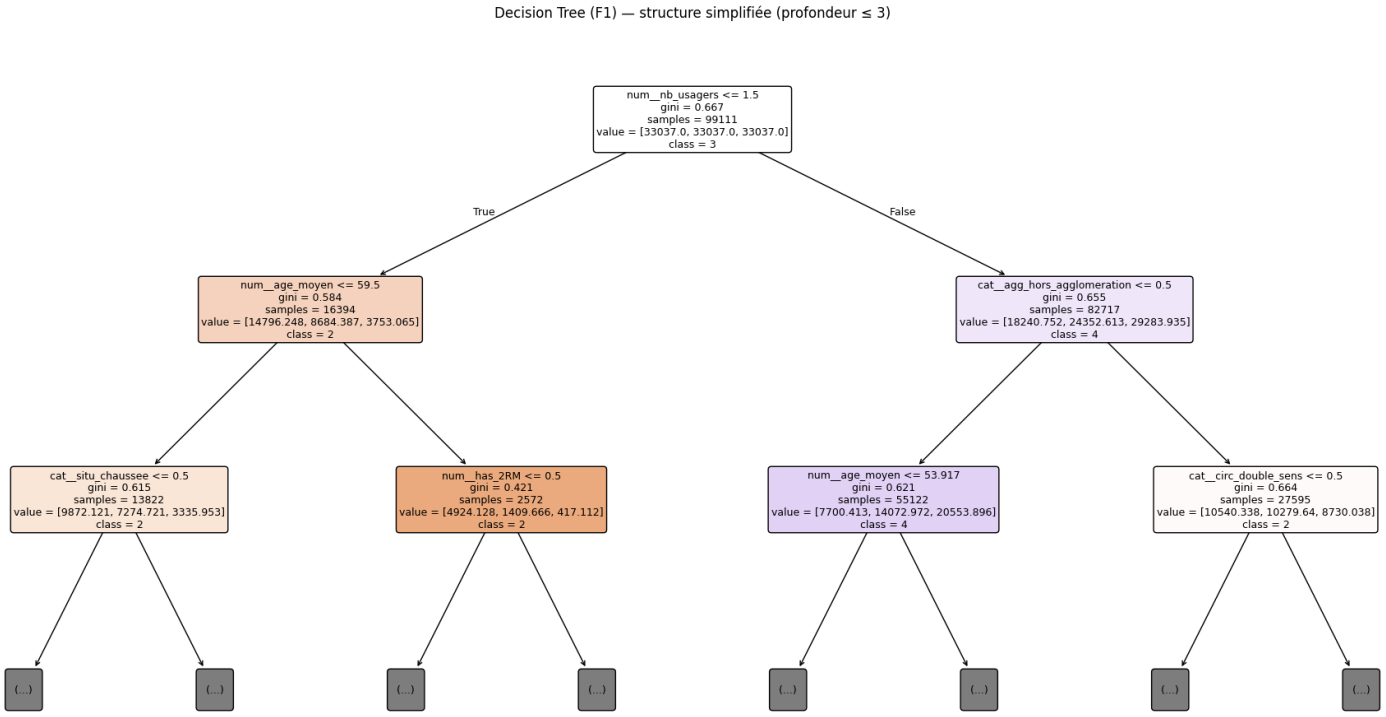


Feature importance Random Forest (F1)

The figure above presents the feature importance derived from the Random Forest trained with meteorological data. As in the configuration without weather data, the average age of users remains the most influential variable, followed by the number of users involved and temporal variables such as day and hour.

A notable point is the appearance of several meteorological variables among the most important features. Atmospheric pressure, temperature, and wind speed thus appear in the top 7 variables, alongside structural and temporal factors (pressure ranked 2nd and temperature ranked 4th).

This observation indicates that the integration of weather data partially modifies the hierarchy of variables used by the model. Meteorological conditions provide complementary information that is taken into account by the Random Forest.



Feature importance Decision Tree (F1)

The visualization of the structure of the Decision Tree (first node) trained with meteorological data shows that weather variables do not appear in the first levels of the tree. The initial decisions mainly rely on structural variables such as the number of users involved and the average age of users.

In order to better analyze the role of meteorological variables within the tree, an additional analysis was conducted by counting their number of occurrences as well as their minimum and average depth in the tree.

	feature_meteo	apparitions	profondeur_min	profondeur_moyenne
0	wind_speed	3	6.0	6.00
1	temperature	7	3.0	5.43
2	pressure	4	5.0	5.75
3	visibility	0	NaN	NaN

The results show that:

- temperature, atmospheric pressure, and wind speed appear several times in the tree,
- However, these variables intervene only at greater depths, typically from level 5 or 6 onward.
- Visibility does not appear in the structure of the selected tree.

These observations indicate that, for the Decision Tree, meteorological variables play a secondary and conditional role: they intervene to refine the decision once the main structural factors have already segmented the data space. Weather is therefore not used as a primary decision criterion, but rather as an adjustment factor in specific configurations.

Consistency between models

The analyses show an overall consistency between the Random Forest and the Decision Tree regarding the consideration of meteorological variables, despite different decision mechanisms.

In the Random Forest, meteorological variables such as temperature, atmospheric pressure, and wind speed appear among the most important features, indicating that they are explicitly and repeatedly integrated into the model's decisions.

In contrast, in the Decision Tree, these same variables do not appear in the first levels of the tree and only emerge at greater depths, once structural variables have already segmented the data space. They are therefore used conditionally, to refine the decision in specific configurations.

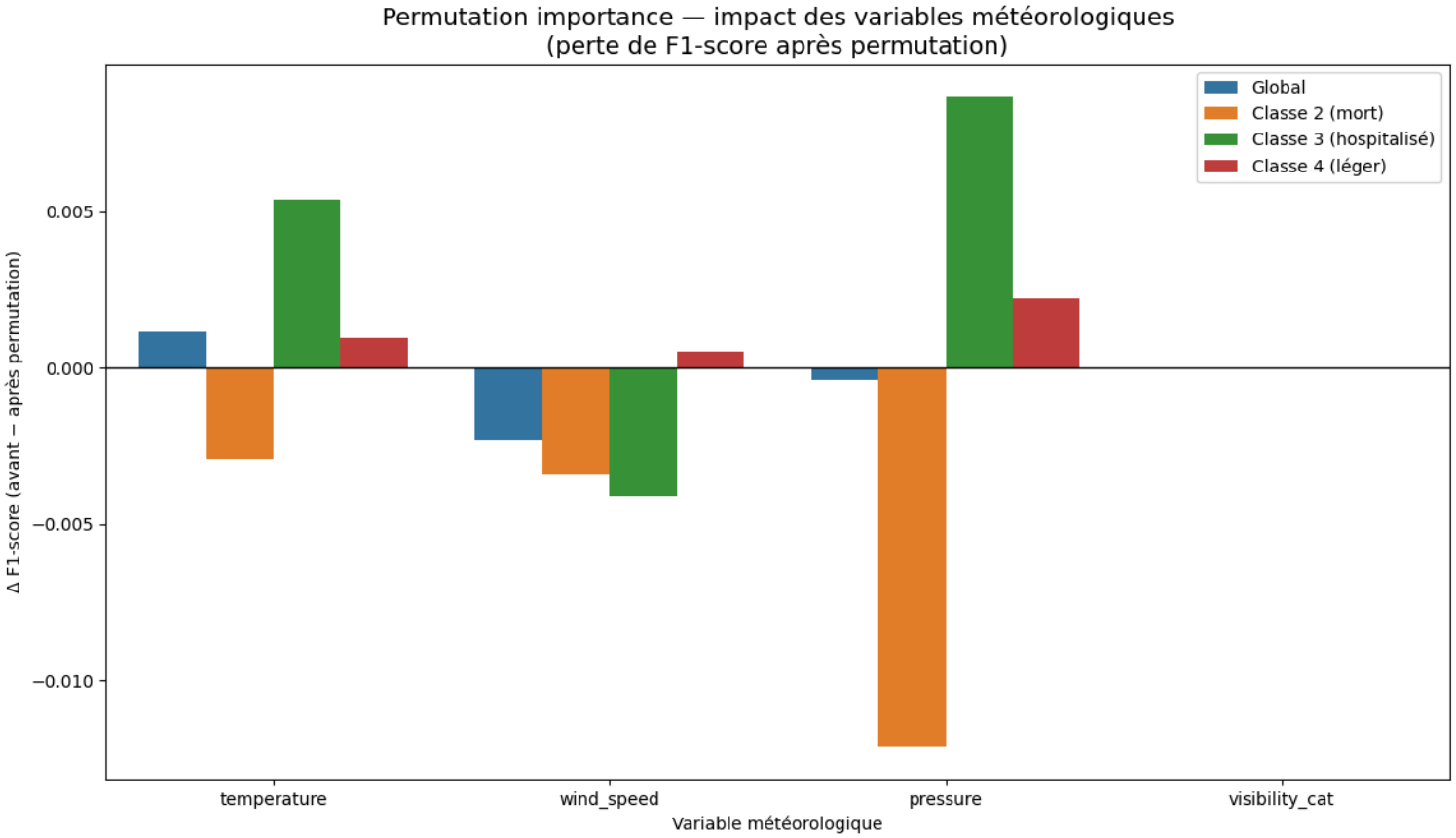
Thus, although weather does not constitute a primary decision criterion, both models converge toward a similar use of these variables: weather is taken into account as a secondary factor, adjusting the prediction once the main determinants related to users and accident context have been identified.

3.3.3 Permutation of meteorological features

A feature permutation analysis was conducted. Permutation consists in randomly shuffling the values of a variable while keeping all other variables unchanged, and then measuring the

variation in model performance. A decrease in performance after permutation indicates that the permuted variable effectively contributes to the prediction.

In this analysis, the impact of permutation is measured through the variation of the F1-score, both globally and by severity class. In the associated plot, a downward-oriented bar corresponds to a decrease in performance after permutation (informative variable, before-after < 0), while an upward-oriented bar indicates a null or non-significant impact. The color of the bars allows distinguishing the global effect from the effect specific to each class.



Interpretation of the results

The results show that meteorological variables have a limited but non-negligible overall impact on model performance. Permuting temperature and atmospheric pressure leads to a slight decrease in the global F1-score, confirming that these variables provide exploitable information for the model.

The class-wise analysis highlights differentiated effects:

- For class 2 (fatal accidents), atmospheric pressure shows the most pronounced impact, with a noticeable decrease in F1-score after permutation, suggesting a

specific role of this variable in predicting the most severe cases.

- For class 3 (hospitalized injury), temperature and pressure contribute positively to model performance.

- For class 4 (slight injury), the impact of meteorological variables remains very weak, indicating that slight severity is not very sensitive to weather conditions.

The visibility variable shows almost no impact, regardless of the class considered, suggesting that it does not provide additional discriminative information in this context.

Conclusion of the permutation analysis

This analysis confirms that meteorological variables play a secondary but real role in predicting accident severity. They do not strongly influence overall model performance, but they can contribute in a targeted way to the prediction of certain classes, particularly the most severe accidents. These results validate the observations derived from feature importance analyses and allow us to conclude that weather mainly acts as a modulating factor, refining the model's decision rather than serving as a primary determinant of accident severity.

Final conclusion

The objective of this work was to evaluate the influence of meteorological conditions on road accident severity, based on classification models trained with and without weather data, followed by an in-depth interpretability analysis. The results show that, in the absence of meteorological data, the models are already able to predict accident severity with non-negligible performance. Variables related to users, their vulnerability (average age), the number of users involved, as well as temporal context and road infrastructure characteristics, constitute the main determinants of severity. Random Forest models stand out for their ability to better capture these complex interactions, particularly for the intermediate class of hospitalized injuries.

The integration of meteorological data leads to limited overall improvements. While a slight gain in accuracy is observed, the macro F1-score remains globally unchanged, indicating that the overall balance between classes is not deeply modified. The class-wise analysis nevertheless reveals more nuanced effects: weather marginally contributes to better differentiation of classes 3 and 4 and modifies the precision/recall trade-off for the fatal accident class, making certain predictions more reliable, but without significantly increasing the detection capability for these rare events.

The interpretability analysis confirms these findings. Meteorological variables such as atmospheric pressure, temperature, and wind speed appear among the important variables in Random Forest models, while they intervene in a more conditional and later manner in Decision Trees. The permutation analysis shows that their impact on performance remains weak to moderate, but non-zero, particularly for predicting the most severe accidents. In

contrast, some variables such as visibility do not show a significant influence in this framework.

Thus, the results allow the research question to be answered in a nuanced way:

Weather does influence accident severity, but in a secondary manner. It does not appear as a determining factor on its own, but rather as a modulating factor, refining the prediction once the main human, contextual, and infrastructural determinants are taken into account.

In conclusion, road accident severity is primarily explained by user characteristics and accident context. Meteorological conditions provide relevant complementary information, but are insufficient to significantly transform model performance or to explain the most severe accidents on their own. Therefore, we must take into account other factors and not merely rely on weather and environmental data in order to get a full picture of the variables of a road accident.